

Indian Statistical Institute

Advanced Statistical Methods II

BSDS (3rd Year)

Solution of Mid-semester Examination held on 10th September, 2026

Problem 1. Let $X \in \{0, 1\}$ with $P(X = 0) = P(X = 1) = \frac{1}{2}$ and suppose

$$E(Y | X = 0) = 0, \quad E(Y | X = 1) = 2.$$

Let $R = 1$ if Y is observed and $R = 0$ otherwise. Consider two mechanisms:

Mechanism A: Suppose $P(R = 1 | X, Y) = q$, $0 < q < 1$.

Mechanism B: Suppose $P(R = 1 | X = 0, Y) = 1$, $P(R = 1 | X = 1, Y) = \frac{1}{2}$.

(i) Classify Mechanisms A and B as MCAR, MAR, or MNAR.

(i) Find the population mean

$$\mu = E(Y).$$

Then, under Mechanism A, compute

$$E(Y | R = 1).$$

(iii) Under Mechanism B, compute

$$E(Y | R = 1).$$

$$[1+(1+1)+2=5]$$

(i) Classification of the two mechanisms.

For Mechanism A,

$$P(R = 1 | X, Y) = q$$

is constant: the probability that Y is observed depends on neither X nor Y . Hence

Mechanism A is MCAR.

For Mechanism B,

$$P(R = 1 | X, Y) = \begin{cases} 1, & X = 0, \\ \frac{1}{2}, & X = 1. \end{cases}$$

The observation probability depends on the observed variable X , but, once X is given, it does not depend on Y . Thus

$$P(R = 1 | X, Y) = P(R = 1 | X),$$

so

Mechanism B is MAR but not MCAR.

[.5 marks for correctly classifying mechanism A, and .5 for correctly classifying mechanism B. The students need not justify why MCAR or MAR.]

(ii) **Population mean and $E(Y | R = 1)$ under Mechanism A.**

$$\begin{aligned} E(Y) &= P(X = 0)E(Y | X = 0) + P(X = 1)E(Y | X = 1) \\ &= \frac{1}{2}(0) + \frac{1}{2}(2) = 1. \end{aligned}$$

Therefore,

$$\boxed{E(Y) = 1.}$$

[If someone proceeds in this direction but could not finally obtain the correct value, give .5 marks.]

Under Mechanism A, $P(R = 1 | X, Y) = q$ is constant, so R is independent of (X, Y) and therefore independent of Y . Hence

$$E(Y | R = 1) = E(Y) = 1.$$

Thus

$$\boxed{E(Y | R = 1) = 1 \quad \text{under Mechanism A.}}$$

(iii) **$E(Y | R = 1)$ under Mechanism B.**

Under Mechanism B,

$$P(R = 1 | X = 0, Y) = 1, \quad P(R = 1 | X = 1, Y) = \frac{1}{2}.$$

Since the missingness probability depends only on X and not on Y once X is given,

$$E(Y | X = x, R = 1) = E(Y | X = x).$$

Now:

$$\begin{aligned} P(R = 1) &= P(R = 1 | X = 0) \cdot P(X = 0) + P(R = 1 | X = 1) \cdot P(X = 1) \\ &= 1 \cdot \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) \cdot \left(\frac{1}{2}\right) \\ &= \frac{3}{4}. \end{aligned}$$

Therefore, by Bayes' rule,

$$P(X = 0 | R = 1) = \frac{P(R = 1 | X = 0)P(X = 0)}{P(R = 1)} = \frac{1 \cdot \frac{1}{2}}{\frac{3}{4}} = \frac{2}{3},$$

and

$$P(X = 1 | R = 1) = \frac{P(R = 1 | X = 1)P(X = 1)}{P(R = 1)} = \frac{\frac{1}{2} \cdot \frac{1}{2}}{\frac{3}{4}} = \frac{1}{3}.$$

Hence, under Mechanism B,

$$\begin{aligned} &E(Y | R = 1) \\ &= E(Y | X = 0, R = 1) \cdot P(X = 0 | R = 1) + E(Y | X = 1, R = 1) \cdot P(X = 1 | R = 1) \\ &= E(Y | X = 0) \cdot \left(\frac{2}{3}\right) + E(Y | X = 1) \cdot \left(\frac{1}{3}\right) \\ &= 0 \cdot \left(\frac{2}{3}\right) + 2 \cdot \left(\frac{1}{3}\right) \\ &= \boxed{\frac{2}{3}}. \end{aligned}$$

[If someone proceeds in the correct direction but could not finally obtain the correct value, give partial marks (while maintaining uniformity across the students). Also, check if there is any conceptual mistake. Evaluate accordingly. As a general scheme: if someone obtained $P(R = 1)$ correctly but not anything else, then give 1 mark.]

Problem 2. Suppose $X_1, \dots, X_n$ arise independently from the two-component Poisson mixture

$$f(x) = \pi \frac{e^{-\lambda_1} \lambda_1^x}{x!} + (1 - \pi) \frac{e^{-\lambda_2} \lambda_2^x}{x!},$$

where $0 < \pi < 1$, $\lambda_1 > 0$, and $\lambda_2 > 0$ are unknown. Introduce the latent indicator

$$Z_i = \begin{cases} 1, & \text{if } X_i \text{ comes from component 1,} \\ 0, & \text{otherwise.} \end{cases}$$

(i) Obtain

$$\tau_i^{(t)} = P(Z_i = 1 \mid X_i = x_i, \theta^{(t)}).$$

(ii) Using the expected complete-data log-likelihood, find an expression for $Q(\theta \mid \theta^{(t)})$, where $\theta^{(t)} = (\pi^{(t)}, \lambda_1^{(t)}, \lambda_2^{(t)})$. Using this, show that the EM updates are

$$\pi^{(t+1)} = \frac{1}{n} \sum_{i=1}^n \tau_i^{(t)},$$

$$\lambda_1^{(t+1)} = \frac{\sum_{i=1}^n \tau_i^{(t)} x_i}{\sum_{i=1}^n \tau_i^{(t)}}, \quad \lambda_2^{(t+1)} = \frac{\sum_{i=1}^n (1 - \tau_i^{(t)}) x_i}{\sum_{i=1}^n (1 - \tau_i^{(t)})}.$$

[2+(1+3)=6]

Let

$$f_j(x) = \frac{e^{-\lambda_j} \lambda_j^x}{x!}, \quad j = 1, 2.$$

The unobserved indicator Z_i records the component from which X_i arose.

(i) At iteration t , let

$$\theta^{(t)} = (\pi^{(t)}, \lambda_1^{(t)}, \lambda_2^{(t)}).$$

By Bayes' rule,

$$\begin{aligned} \tau_i^{(t)} &= P(Z_i = 1 \mid X_i = x_i, \theta^{(t)}) \\ &= \frac{P(Z_i = 1 \mid \theta^{(t)}) P(X_i = x_i \mid Z_i = 1, \theta^{(t)})}{P(X_i = x_i \mid \theta^{(t)})} \\ &= \frac{\pi^{(t)} e^{-\lambda_1^{(t)}} (\lambda_1^{(t)})^{x_i} / x_i!}{\pi^{(t)} e^{-\lambda_1^{(t)}} (\lambda_1^{(t)})^{x_i} / x_i! + (1 - \pi^{(t)}) e^{-\lambda_2^{(t)}} (\lambda_2^{(t)})^{x_i} / x_i!} \\ &= \frac{\pi^{(t)} e^{-\lambda_1^{(t)}} (\lambda_1^{(t)})^{x_i}}{\pi^{(t)} e^{-\lambda_1^{(t)}} (\lambda_1^{(t)})^{x_i} + (1 - \pi^{(t)}) e^{-\lambda_2^{(t)}} (\lambda_2^{(t)})^{x_i}}. \end{aligned}$$

[If someone proceeds in the correct direction and applies Bayes Rule but could not finally obtain the correct expression, give 1 mark.]

(ii) If the Z_i 's were observed, the complete-data likelihood would be

$$L_c(\theta) = \prod_{i=1}^n [\pi f_1(x_i)]^{Z_i} [(1 - \pi) f_2(x_i)]^{1-Z_i}.$$

Hence, apart from terms independent of the parameters,

$$\begin{aligned}\ell_c(\theta) &= \sum_{i=1}^n \left[Z_i \log \pi + (1 - Z_i) \log(1 - \pi) \right] \\ &\quad + \sum_{i=1}^n Z_i \{-\lambda_1 + x_i \log \lambda_1\} \\ &\quad + \sum_{i=1}^n (1 - Z_i) \{-\lambda_2 + x_i \log \lambda_2\} + \text{constant}.\end{aligned}$$

In the E-step, replace Z_i by its conditional expectation $\tau_i^{(t)}$. Thus

$$\begin{aligned}Q(\theta \mid \theta^{(t)}) &= \sum_{i=1}^n \left[\tau_i^{(t)} \log \pi + (1 - \tau_i^{(t)}) \log(1 - \pi) \right] \\ &\quad + \sum_{i=1}^n \tau_i^{(t)} \{-\lambda_1 + x_i \log \lambda_1\} \\ &\quad + \sum_{i=1}^n (1 - \tau_i^{(t)}) \{-\lambda_2 + x_i \log \lambda_2\} + \text{constant}.\end{aligned}$$

For π ,

$$\frac{\partial Q}{\partial \pi} = \sum_{i=1}^n \left[\frac{\tau_i^{(t)}}{\pi} - \frac{1 - \tau_i^{(t)}}{1 - \pi} \right].$$

Setting this equal to zero gives

$$\sum_i \tau_i^{(t)} (1 - \pi) = \pi \sum_i (1 - \tau_i^{(t)}),$$

and therefore

$$\boxed{\pi^{(t+1)} = \frac{1}{n} \sum_{i=1}^n \tau_i^{(t)}}.$$

For λ_1 ,

$$\frac{\partial Q}{\partial \lambda_1} = \sum_{i=1}^n \tau_i^{(t)} \left(-1 + \frac{x_i}{\lambda_1} \right).$$

Setting this equal to zero gives

$$-\sum_i \tau_i^{(t)} + \frac{1}{\lambda_1} \sum_i \tau_i^{(t)} x_i = 0,$$

so

$$\boxed{\lambda_1^{(t+1)} = \frac{\sum_{i=1}^n \tau_i^{(t)} x_i}{\sum_{i=1}^n \tau_i^{(t)}}}.$$

Doing the exact same calculation for λ_2 gives

$$\boxed{\lambda_2^{(t+1)} = \frac{\sum_{i=1}^n (1 - \tau_i^{(t)}) x_i}{\sum_{i=1}^n (1 - \tau_i^{(t)})}}.$$

[Each of the three EM updates carries 1 mark.]

Problem 3. Let

$$F(\theta) = \sum_{i=1}^n |y_i - \theta|, \quad \theta \in \mathbb{R},$$

and suppose that $y_i \neq \theta^{(t)}$ for every i at the current iterate.

(i) Using the inequality

$$|u| \leq \frac{u^2}{2|u_0|} + \frac{|u_0|}{2}, \quad u_0 \neq 0,$$

or otherwise, construct an MM surrogate for $F(\theta)$ at the current iterate $\theta^{(t)}$.

(ii) Show that minimizing this surrogate gives the following update.

$$\theta^{(t+1)} = \frac{\sum_{i=1}^n w_i^{(t)} y_i}{\sum_{i=1}^n w_i^{(t)}}, \quad w_i^{(t)} = \frac{1}{|y_i - \theta^{(t)}|}.$$

[2+3=5]

(i) Construction of the MM surrogate.

For the i th term, apply the given inequality with

$$u = y_i - \theta, \quad u_0 = y_i - \theta^{(t)}.$$

Because $y_i \neq \theta^{(t)}$, we have $u_0 \neq 0$, and hence

$$|y_i - \theta| \leq \frac{(y_i - \theta)^2}{2|y_i - \theta^{(t)}|} + \frac{|y_i - \theta^{(t)}|}{2}.$$

Summing this inequality over $i = 1, \dots, n$ gives

$$F(\theta) \leq G(\theta \mid \theta^{(t)}),$$

where

$$G(\theta \mid \theta^{(t)}) = \frac{1}{2} \sum_{i=1}^n \frac{(y_i - \theta)^2}{|y_i - \theta^{(t)}|} + \frac{1}{2} \sum_{i=1}^n |y_i - \theta^{(t)}|.$$

At $\theta = \theta^{(t)}$, equality holds term by term in the given inequality, and therefore

$$G(\theta^{(t)} \mid \theta^{(t)}) = F(\theta^{(t)}).$$

Thus $G(\theta \mid \theta^{(t)})$ is a valid MM surrogate for $F(\theta)$.

[If someone arrives at the correct $G(\theta \mid \theta^{(t)})$ function but does not justify why it is a valid MM surrogate for $F(\theta)$ then deduct 1 mark.]

(ii) Minimization of the surrogate.

The second sum in $G(\theta \mid \theta^{(t)})$ does not depend on θ . Therefore, minimizing the surrogate is equivalent to minimizing

$$\sum_{i=1}^n \frac{(y_i - \theta)^2}{|y_i - \theta^{(t)}|} = \sum_{i=1}^n w_i^{(t)} (y_i - \theta)^2,$$

where

$$w_i^{(t)} = \frac{1}{|y_i - \theta^{(t)}|}.$$

Differentiate with respect to θ :

$$\frac{d}{d\theta} \sum_{i=1}^n w_i^{(t)} (y_i - \theta)^2 = -2 \sum_{i=1}^n w_i^{(t)} (y_i - \theta).$$

Setting this derivative equal to zero gives

$$\sum_{i=1}^n w_i^{(t)} y_i - \theta \sum_{i=1}^n w_i^{(t)} = 0.$$

Hence the unique minimizer is

$$\theta^{(t+1)} = \frac{\sum_{i=1}^n w_i^{(t)} y_i}{\sum_{i=1}^n w_i^{(t)}}, \quad w_i^{(t)} = \frac{1}{|y_i - \theta^{(t)}|}.$$

In words, one MM step replaces the LAD objective by a weighted least-squares surrogate, with observations closer to the current iterate receiving larger weights.

Problem 4. Consider the following survival data observed for six individuals

$$(2, 1), \quad (3, 0), \quad (4, 1), \quad (4, 1), \quad (6, 0), \quad (7, 1),$$

where the first coordinate is the observed time and the second coordinate is

$$\delta = \begin{cases} 1, & \text{failure,} \\ 0, & \text{right censoring.} \end{cases}$$

- (i) Identify the distinct event times and, at each event time, determine the corresponding risk-set size n_j and number of failures d_j .
- (ii) Compute the Kaplan–Meier estimator $\hat{S}(t)$ at all distinct event times.
- (iii) Is the Kaplan–Meier median survival time estimable from these data? If yes, give it; if not, explain why.

$$[3+2+1=6]$$

(i) Event times and risk sets.

The distinct failure/event times are

$$\boxed{2, 4, 7.}$$

At time 2, all six individuals are still under observation immediately before time 2. Hence

$$n_1 = 6, \quad d_1 = 1.$$

After the failure at time 2, five remain. One individual is censored at time 3, so immediately before time 4 there are four individuals at risk. Two fail at time 4. Hence

$$n_2 = 4, \quad d_2 = 2.$$

After the two failures at time 4, two individuals remain. One is censored at time 6, leaving one individual at risk immediately before time 7. That individual fails at time 7. Hence

$$n_3 = 1, \quad d_3 = 1.$$

Thus

event time t_j	2	4	7
n_j	6	4	1
d_j	1	2	1

[Each of the six numbers in the box just above carries .5 marks. Evaluate accordingly.]

(ii) Kaplan–Meier estimator.

The Kaplan–Meier estimator is

$$\hat{S}(t) = \prod_{t_j \leq t} \left(1 - \frac{d_j}{n_j}\right).$$

At the first event time,

$$\hat{S}(2) = 1 - \frac{1}{6} = \frac{5}{6}.$$

At the second event time,

$$\hat{S}(4) = \left(1 - \frac{1}{6}\right) \left(1 - \frac{2}{4}\right) = \frac{5}{6} \cdot \frac{1}{2} = \frac{5}{12}.$$

At the final event time,

$$\hat{S}(7) = \frac{5}{12} \left(1 - \frac{1}{1}\right) = 0.$$

Hence the estimated survival curve is

$$\hat{S}(t) = \begin{cases} 1, & t < 2, \\ \frac{5}{6}, & 2 \leq t < 4, \\ \frac{5}{12}, & 4 \leq t < 7, \\ 0, & t \geq 7. \end{cases}$$

[Each of the four numbers in the box just above carries .5 marks. If someone gives the values at the time points 2, 4 and 7 then also give full marks. Evaluate accordingly.]

(iii) Median survival time.

The Kaplan–Meier median survival time is conventionally defined as

$$\hat{m} = \inf\{t : \hat{S}(t) \leq 1/2\}.$$

We have

$$\hat{S}(2) = \frac{5}{6} > \frac{1}{2},$$

while

$$\hat{S}(4) = \frac{5}{12} < \frac{1}{2}.$$

Therefore the curve first reaches or falls below $1/2$ at time 4, so

$$\boxed{\hat{m} = 4.}$$

Thus the Kaplan–Meier median is estimable from these data.

[If someone just mentions that yes it is estimable, but does not derive the estimate, then no marks should be given.]